PROJECT PROPOSAL

SMART GREEN TECHNOLOGY

Solar-Powered Smart Street Lighting System

Adaptive, energy-independent street lighting using solar power and motion-based dimming

Submitted by:

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1. Abstract

This project proposes a solar-powered smart street lighting system that combines photovoltaic panels for energy independence with motion sensors and ambient-light sensors for adaptive brightness control. Streetlights dim during periods of low activity and brighten when motion is detected, reducing energy consumption while maintaining safety, and the system reports status/faults remotely for easier maintenance.

2. Introduction

Conventional street lighting runs at full brightness throughout the night regardless of pedestrian or vehicle activity, consuming grid electricity continuously and requiring manual fault detection. Solar-powered LED street lights with adaptive control can operate independently of the grid while further reducing energy use through motion-responsive dimming, making them well suited for both urban and rural deployment.

3. Problem Statement

Grid-connected street lighting consumes significant electricity, contributes to carbon emissions, and depends on centralized power availability; faults such as a bulb outage often go unnoticed until reported by the public. There is a need for an energy-independent, adaptive lighting solution with remote fault monitoring.

4. Objectives

- To design a solar-powered LED street light unit with battery storage for night-time operation.
- To implement motion and ambient-light sensing for adaptive brightness control.
- To develop a remote monitoring system for tracking battery status and detecting faults.
- To evaluate energy savings compared to conventional always-on street lighting.
- To assess reliability across varying weather/seasonal conditions.

5. Literature Survey / Existing System

Solar street lighting has been widely studied as a renewable alternative to grid-powered lighting, with research focusing on optimal panel/battery sizing for reliable night-long operation. Motion-responsive dimming (reducing brightness to a baseline level and brightening on detected motion) has been shown in various studies to substantially cut energy consumption without compromising perceived safety. This project integrates both aspects along with remote fault-reporting for practical municipal deployment.

6. Proposed System

The proposed system uses a solar panel to charge a battery during the day, powering an LED light at night. A PIR motion sensor and LDR (light-dependent resistor) allow the light to dim to a low baseline level when no activity is detected and brighten when motion is sensed nearby. A microcontroller with a wireless module periodically reports battery charge level and light status to a central monitoring dashboard, flagging any unit that fails to report or shows abnormal battery drain.

6.1 Methodology

- Sizing and selection of solar panel, battery, and LED components for reliable night-long operation.
- Circuit design integrating PIR motion sensor and LDR for adaptive brightness control.
- Microcontroller programming for charge/discharge management and adaptive lighting logic.
- Wireless status reporting setup (battery level, fault flags) to a central dashboard.
- Dashboard development for monitoring multiple light units.
- Field/prototype testing across different activity and weather conditions.

7. Tools and Technologies

- Solar panel and charge controller
- Li-ion/Lead-acid battery for storage
- PIR motion sensor and LDR sensor
- ESP32/Arduino microcontroller
- LoRa/GSM module for remote reporting
- Web dashboard (React/Flask) for monitoring

8. System Requirements

8.1 Hardware Requirements

- Solar panel
- Charge controller
- Rechargeable battery
- High-efficiency LED light
- PIR motion sensor
- LDR sensor
- ESP32/Arduino microcontroller
- LoRa/GSM communication module

8.2 Software Requirements

- Arduino IDE
- Backend framework (Flask/Node.js) for the dashboard
- Database (Firebase/MySQL) for status logs

9. Project Timeline

Phase	Activity	Duration
Phase 1	Component sizing and requirement study	1 week
Phase 2	Circuit design and prototype assembly	3 weeks

Phase	Activity	Duration
Phase 3	Adaptive lighting logic development	2 weeks
Phase 4	Remote monitoring and fault-reporting integration	3 weeks
Phase 5	Dashboard development	2 weeks
Phase 6	Field testing and documentation	2 weeks

10. Expected Outcomes

- A working solar-powered street light prototype with adaptive motion-based dimming.
- A remote monitoring dashboard tracking battery health and fault status across units.
- Quantified energy savings compared to always-on conventional lighting.

11. Advantages

- Operates independently of the grid, reducing electricity costs and emissions
- Extends battery/component life through adaptive dimming
- Enables proactive maintenance via remote fault detection
- Suitable for both urban and off-grid rural deployment

12. Future Scope

The system can be extended with mesh networking between light units for coordinated brightness along a street, integration with traffic/pedestrian density sensors for smarter adaptive control, and predictive maintenance models based on battery degradation trends.

13. References

[1] Research articles on solar-powered adaptive street lighting (refer to IEEE/Elsevier Renewable Energy journals for latest citations).

[2] LoRa Alliance Technical Documentation, lora-alliance.org

[3] Studies on PIR-based adaptive street lighting energy savings.

[4] Solar panel and battery sizing guidelines for off-grid lighting systems.